

AE646 Stage 3 (Final) Presentation

# Optimising a Fourier Neural Operator for Parametric Darcy Flow

Recipe, architecture, data, resolution, latency, and comparison with a finite-volume solver

Team: PINNacles (Aryamann Srivastava, Varun Sathaye, Atishay Jain, Vedant S. Tiwari)

# Recap and Stage 3 Questions

Learning the operator  $\kappa \rightarrow u$  of  $-\text{div}(\kappa \text{grad } u) = 1$ , on real PDEBench data

## Where Stage 2 ended

- Real PDEBench pipeline: 900 / 100 / 200 split at 64x64, EDA, preprocessing and baseline ablations
- MLP (42 M): 0.082 mean rel. L2; FNO (4.7 M): 0.052
- Speed-up claim vs an FDM solver (1030 ms)

## Stage 3 questions

- How much of the FNO error was optimisation, and how much capacity?
- Which architecture choices matter? How much data? Which resolution?
- What does inference cost, and how does the FNO compare with a proper finite-volume solver?
- What does the residual error mean physically?

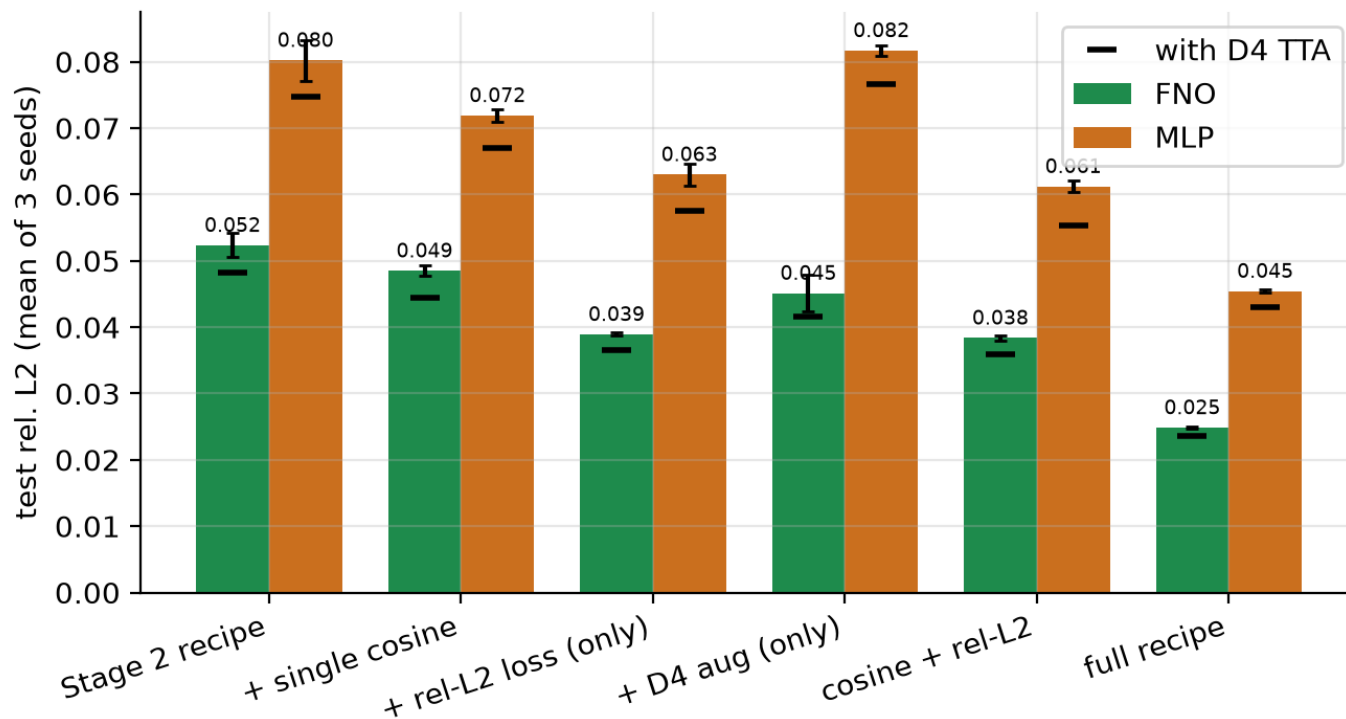
# Experimental Protocol

Same split as Stage 2; every choice made on validation data only

- Data: 900 train / 100 val / 200 test real fields (seed 42); 8,800 further real samples used only for the data-scaling study
- Metric: per-sample relative L2 error in physical units, mean over test samples; test set evaluated once per run
- Reference step budget = Stage 2 (100 epochs = 5,700 steps, batch 16); seeds 0-4; sd over seeds reported
- Symmetry: data are D4-invariant; augmentation applied at 128x128 before sub-sampling (valid), test-time averaging over {identity, transpose}
- Baseline given the same effort: MLP tuned on the same footing (capacity sweep, validation selection)
- ~200 training runs on one shared GPU workstation; all configs and metrics stored per run

# Finding 1: The Training Recipe Limited Stage 2

Same models, same 5,700 steps, 3 seeds - only the recipe changes

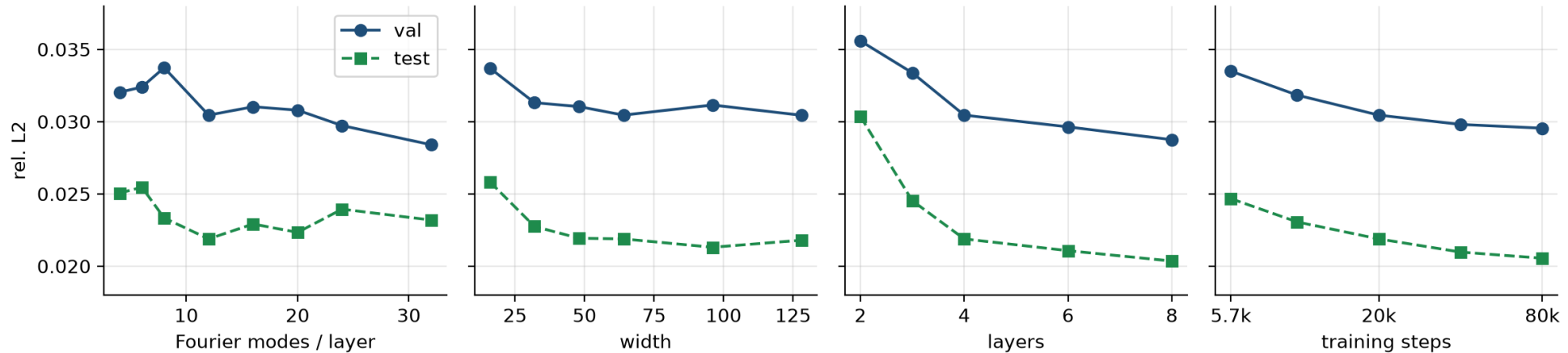


## Result

- Stage 2 recipe: MSE loss; cosine LR restarted every 200 steps
- Relative-L2 loss: -26%. Single cosine: -7%. D4 augmentation on top: -35%
- FNO 0.0524  $\rightarrow$  0.0249 (2.1x lower); MLP 0.0802  $\rightarrow$  0.0454
- Gains interact: D4 alone does not help under the old recipe

# Finding 2: Architecture Choices Matter Little

One-factor sweeps around (12 modes, width 64, 4 layers); selection on validation error



## What we learned

- Modes 4 to 32 (0.5 M to 34 M parameters): only 0.022-0.025 test error; width  $\geq 32$  saturates
- Depth and training length help slowly: 2  $\rightarrow$  8 layers 0.030  $\rightarrow$  0.020; 5.7k  $\rightarrow$  80k steps 0.0247  $\rightarrow$  0.0205
- Learning rate, weight decay, batch size:  $< 0.001$  effect. Validation picks deeper, not wider: (12, 64, 8), 9.5 M

# Final Models: FNO vs Tuned MLP

60,000 steps, full recipe, 5 seeds, 200 held-out test fields

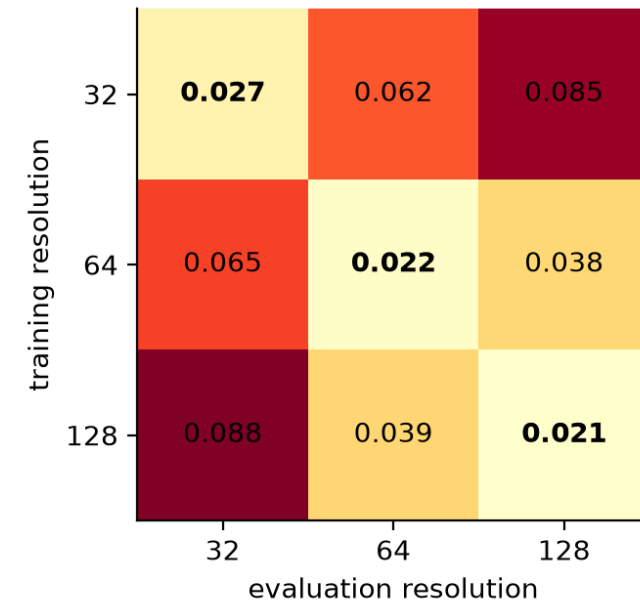
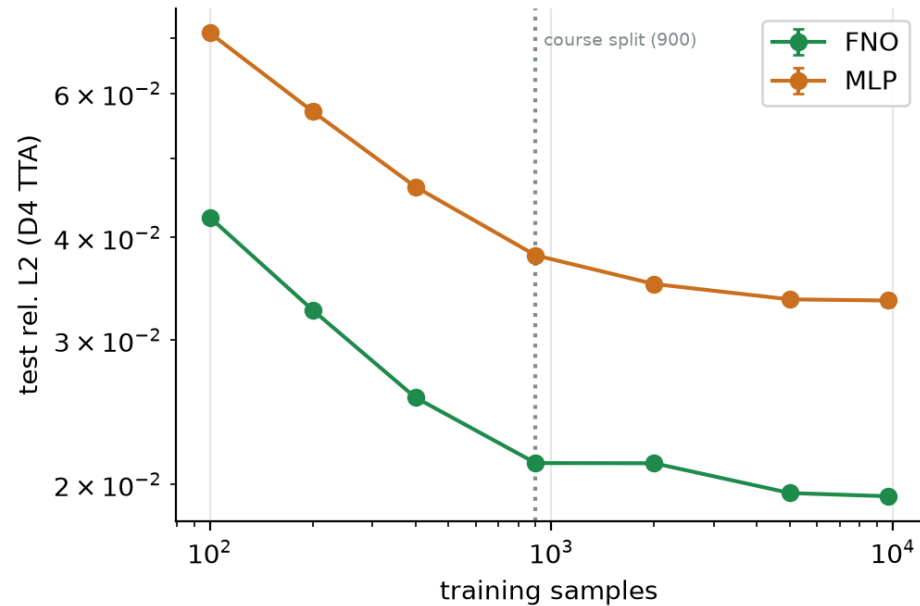
| Model                      | Params       | Mean rel. L2            | Median        | With TTA      |
|----------------------------|--------------|-------------------------|---------------|---------------|
| Stage 2 MLP                | 42.0 M       | 0.0820                  | 0.0695        | -             |
| Stage 2 FNO                | 4.7 M        | 0.0521                  | 0.0398        | -             |
| Final MLP (3x4096)         | 100.7 M      | 0.0415 +- 0.0009        | 0.0272        | 0.0386        |
| <b>Final FNO (w64, L8)</b> | <b>9.5 M</b> | <b>0.0202 +- 0.0004</b> | <b>0.0070</b> | <b>0.0200</b> |

## Take-aways

- FNO error 2.05x lower than an equally tuned MLP (Stage 2 gap: 1.6x) with 10.6x fewer parameters; 2.6x lower than the Stage 2 FNO
- Median FNO error is 0.7%: the mean is dominated by a small tail (next slides)
- Training the final FNO: about 32 minutes on one shared GPU

# Finding 3: Data Scaling and Resolution

FNO vs MLP with 100 to 9,700 samples; train-vs-evaluation resolution matrix



- FNO with 400 samples beats the MLP with 9,700; FNO plateaus near 0.0195 for  $\geq 5,000$  samples
- Zero-shot transfer 64  $\rightarrow$  128: 0.038 (native 128: 0.021); 64  $\rightarrow$  32: 0.065: useful, but 1.7-3x worse than native resolution

# Finding 4: Comparison with a Finite-Volume Solver

Correction to Stage 2: the old FDM reference was inefficient - the 1443x / 7740x speed-ups are withdrawn

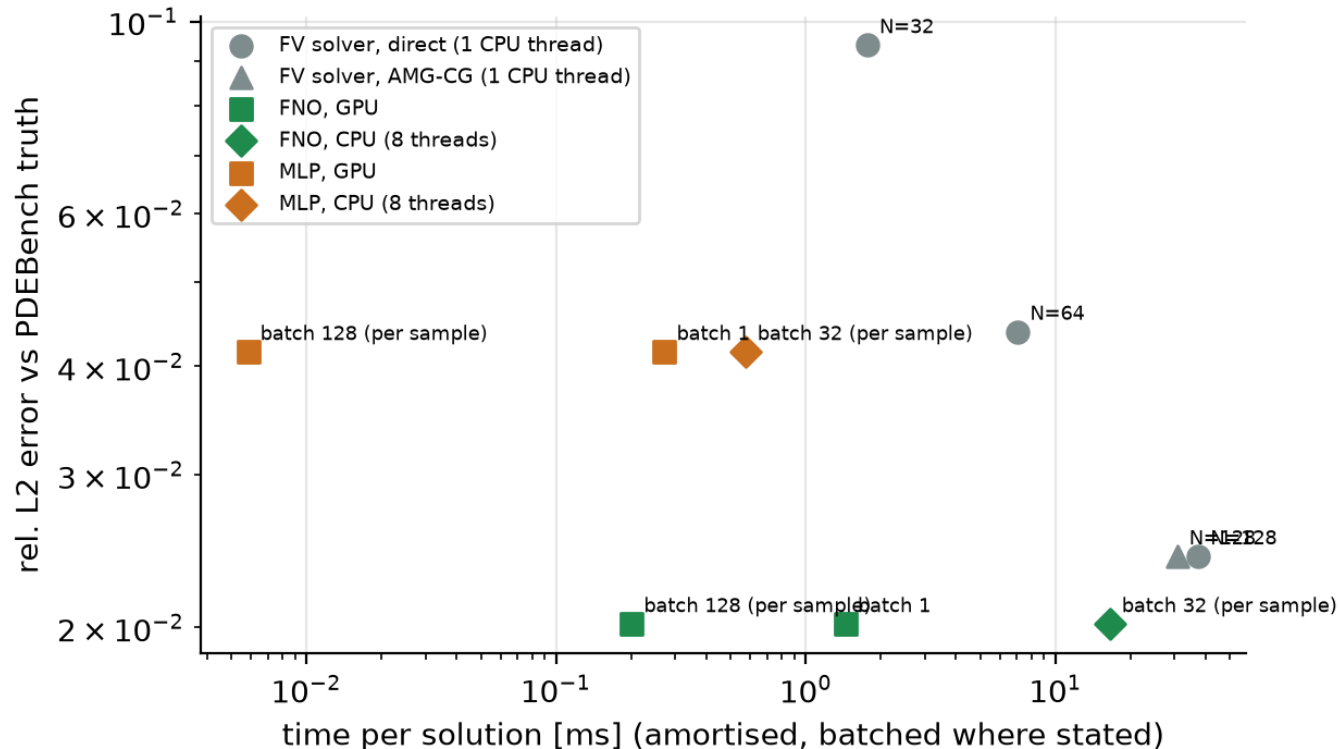
| Grid                      | Calibrated FV           | Error vs truth | Direct LU           | AMG-CG                   |
|---------------------------|-------------------------|----------------|---------------------|--------------------------|
| 32 x 32                   | harmonic, b = 0.7       | 0.094          | 1.8 ms              | 5.5 ms                   |
| 64 x 64                   | harmonic, b = 0.8       | 0.044          | 7.1 ms              | 11.6 ms                  |
| 128 x 128                 | arithmetic, b = 1.05    | 0.024          | 37.4 ms             | 30.7 ms                  |
| <b>FNO, 64 x 64 (GPU)</b> | <b>9.5 M parameters</b> | <b>0.020</b>   | <b>1.4 ms (b=1)</b> | <b>0.20 ms (batched)</b> |

## How to read it

- Vectorised solver; boundary distance b calibrated on validation data (PDEBench's truth behaves as if the Dirichlet boundary were ~1 cell outside the array)
- At equal resolution the FNO agrees with the data 2.2x better; partly because it learns the generator's own discretisation
- Single CPU thread: FNO 37 ms at batch 1 - NOT faster than the  $64^2$  solver (7 ms); the gain is GPU batching

# Inference Cost: Error vs Time per Solution

CUDA-event timing on an idle Blackwell GPU; CPU numbers from a shared host (indicative)

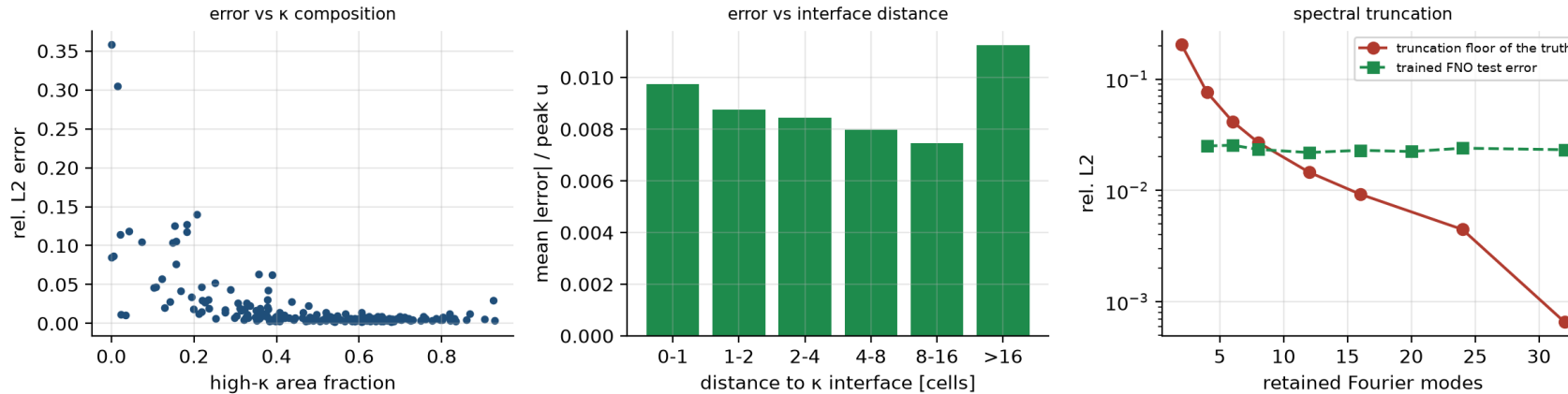


## Latency at 64 x 64

- FNO (9.5 M): 1.44 ms at batch 1, 0.20 ms per sample at batch 128
- 5x (batch 1) to 35x (batched) faster than the  $64^2$  solver, and 26-187x faster than the  $128^2$  solver
- MLP: 0.006 ms/sample but 2x worse accuracy
- TF32 gives no gain; bfloat16 unsupported for complex FFT
- Break-even vs the  $128^2$  solver:  $\sim 5e4$  solves

# Physical Interpretation

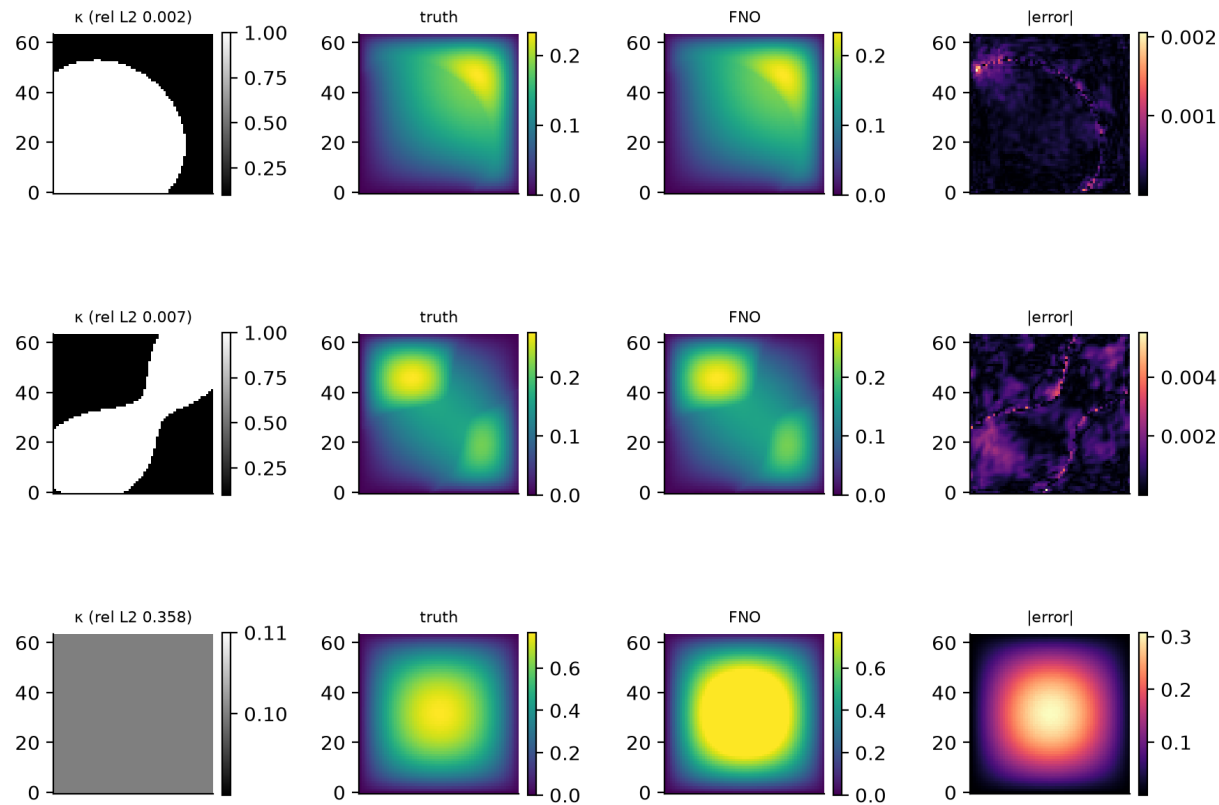
Final FNO on the 200 test fields



- Physics respected: PDE residual within 6% of the truth's own; mean pressure to 1.7%; boundary ring ratio 0.1157 vs 0.1159;  $1.5e-5$  of pixels negative
- Error is NOT concentrated at interfaces (0.0098 next to them vs 0.0075 at 8-16 cells); largest in large low-kappa regions with high pressure ( $r = 0.61$  with peak  $u$ )
- Spectral: truth needs  $\sim 12$  modes (1.5% truncation error), but a 4-mode FNO still reaches 2.5%: the bypass, nonlinearity and depth regenerate high frequencies

# Best, Median and Worst Test Sample

kappa | truth | FNO | absolute error



## Reading the figure

- Best 0.002, median 0.007
- Worst 0.358: uniform kappa = 0.1; correct shape, saturated amplitude
- The two uniform fields have errors 0.358 and 0.085; without them the mean falls from 2.0% to 1.8%
- Since  $u \sim 1/\kappa$  for uniform kappa, a scale-normalised formulation could remove this failure (not tested)

# Limitations and Honest Notes

What the results do and do not show

- One PDE, one dataset, binary kappa,  $f = 1$ ; the plateau and scaling laws may not transfer
- Top architectures differ by less than seed noise; single seed per sweep point (final models: 5 seeds)
- Stage 2 absolute errors reflected a sub-optimal recipe (same for both models, so the comparison was fair); Stage 2 speed-up figures withdrawn
- Solver was calibrated to the dataset: the comparison measures agreement with PDEBench, not fidelity to the continuous PDE; single-thread, uncompiled solver
- Timings on a shared machine: CPU numbers are indicative; GPU timing on an idle GPU

# Summary

## Recipe

Relative-L2 loss + single cosine + D4 augmentation halve the FNO error at equal cost (0.052 -> 0.025)

## Final

FNO 9.5 M: 0.0202 +- 0.0004 vs tuned MLP 100.7 M: 0.0415 +- 0.0009 (5 seeds): 2.05x, 10.6x fewer parameters

## Limits

Insensitive to modes/width; plateaus near 2% for  $\geq 5,000$  samples; zero-shot resolution transfer is moderate

## Solver

2.2x more accurate than a calibrated FV solver at 64x64 and 5-35x faster on GPU; not faster on one CPU core

## Physics

Respects BC, positivity, discrete PDE; residual error sits in uniform / low-contrast, high-amplitude fields

Thank you - Questions welcome